
a Expert chemical signatures (amino-acid enrichment)
Sulfur /metal
Basic
Acidic
Polar uncharged
Hydrophobic
Aromatic
G/P
DNA contact enrichment
Structural scaffold
Mixed
DNA-interface
e5 (4%)
e3 (17%)
e6 (10%)
e0 (17%)
e7 (11%)
e2 (18%)
e1 (14%)
e4 (9%)
-0.66
-0.46
-0.37
+0.16
+0.20
+0.25
+0.25
+0.31
C M H K R D E Q N S T A V L I F Y W G P
Depleted
Enriched
Enrichment (log2, family-controlled)
b Expert structural-role signatures
Zn coord.
Base contact
Backbone
Buried core
Helix
Loop/linker
Surface
Enrichment (log2 odds)
Structural scaffolds
Mixed
DNA interface
|effect|
0.4
0.9
1.7
c Expert co-routing structure (top-2 routing)
Co-routing freq. (%)
Enriched pairs (red outline)
log2 obs/exp
e0 – e1 major-groove readout + backbone interface +1.79
e4 – e7 His–Zn scaffold + minor-groove reader +1.52
e2 – e3 base readout + helical scaffold +1.32
e5 – e6 buried/loop scaffold + Cys–Zn scaffold +0.84
d Case study: C2H2 zinc finger
Cys
Cys
Recognition helix
His
His
e6
e2 / e1
e4
e6 Cys–Zn scaffold
e4 His–Zn scaffold
e2 base readout (β-element)
e1 backbone interface
n = 5,475 fingers
Recognition-helix position (canonical C2H2 numbering)
e Case study: bHLH
Basic DNA-binding region
Helix 1 / HLH (dimerisation)
e0
e3
e0 major-groove readout
e3 helical / buried scaffold
n = 21 domains
Domain position (0 = end of basic region)
f Cross-family reuse of biochemical programs
e1: backbone interface
e3: helical / buried scaffold
e6: Cys–Zn scaffold
e7: minor-groove / β-element
Program consistency
C2H2
bHLH
bZIP
HD
ETS
n/a